PRATHIK ANAND KRISHNAN

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CAREER SUMMARY

- Skilled Backend Engineer with 2.5 years of experience in high-performance systems including FinTech.
- Strong understanding of Crypto HFT systems, with hands-on experience in building low-latency C++ trading system algorithms.
- Experienced in developing real-time market data pipelines, and OEMS architectures for both Centralized and Decentralized exchanges.
- Specializing in Modern C++ (C++17) and Python. Currently learning Rust.
- Proficient in design patterns (UMLs), multithreading, asynchronous programming and debugging complex legacy C++ systems.
- Strong team collaborator with experience in Agile delivery and writing robust, maintainable code.

SKILLS & TECHNOLOGIES

Technical Skills & Competencies

- Web & Backend Microservice Development
- Software development life cycle
- Agile delivery and Scrum processes
- Build & Integration toolchain, CI/CD
- GUI & Frontend
- Testing & Automation
- Databases
- Networking & Protocols

Technologies & Tools

- C++11,17, Python, Rust, Shell Scripting
- Linux, Windows
- Git, JIRA, MIRO board, Figma
- GCC, MSVC, Meson, Conan, Azure pipelines
- Vue, React, JavaScript, MFC
- GoogleTests, Catch2, PyTest, Python Scripts, TDD
- PostgreSQL, Redis, MySQL
- REST APIs, Proto-buffs, JSON-RPC & WebSockets

WORK EXPERIENCE

BACKEND ENGINEER, Consultant – Remote — 10/2024 – Present

GoQuant – Miami, FL, USA

- Designed and implemented a low-latency OEMS in C++17/Python, integrating REST and WebSocket endpoints for order execution and market data.
- Designed and integrated IOC, OCO, GTT and DAY order algorithms with asynchronous order monitoring and thread-safe updates using mutexes.
- Developed DEX adapters for Uniswap v4, Raydium, Orca and Hyperliquid enabling real-time trade execution, position tracking, and AMM simulations.
- Built off-chain AMM and Blockchain smart contract interaction layer for Uniswap v4 and Orca DEX, supporting on-chain swaps via off-chain transaction signing.
- Implemented WebSocket server (C++17, Drogon) for multi-client symbol subscriptions and live order book streaming.
- Utilized Flatbuffers for high-performance market-data serialization, reducing overall system latency.
- Developed automated builds and testing scripts using Meson, Conan, and Shell scripts achieving end-to-end CI/CD integration and pipeline automation.
- Enhanced testing reliability by using randomized seed unit tests in C++20 and PyTest for stress testing.

SOFTWARE DEVELOPER – G3 — 04/2023 – 10/2024

ARUP – Hyderabad, India

- Developed a C++20 static library Adapter API using the Builder Pattern, deployed as a NuGet package for seamless integration between proprietary applications.
- Applied Test-Driven Development (TDD) using Google Test frameworks, achieving 96% code coverage in SDK and integrated SonarQube and CodeCov for continuous quality checks.

- Wrote Shell scripts for Azure Pipelines to automate build, deployment, and testing workflows.
- Created UML diagrams on MIRO boards for low-level and high-level system architecture visualization.
- Participated in pair programming and Tech-spikes to address technical debt.
- Built a PostHog-based event logging system via REST API, enabling real-time analytics and feature flag management within the application.

SOFTWARE DEVELOPER – G2 11/2022 – 04/2023

ARUP – Hyderabad, India

- Improved UI responsiveness by offloading resource-heavy operations to a separate std::thread, while using mutexes to preserve data integrity and thread safety.
- Wrote units tests for code, achieving 86% code coverage for the project. Implemented hot-fixes and bug fixes identified during ensemble tests and black-box testing.
- Implemented MFC UI features and backend logics to effectively execute synchronization initiative.
- I was also hosting Monthly Software Development Meets (OSM) at ARUP, creating a platform for knowledge sharing and was contributing to our team's technical growth.

Relevant Internships

- MATLAB developer | CSIR – National Geophysical Research institute, Hyderabad | June 2019 – Aug 2019
- Post-Graduate Engineer | ATKINS | India | Jan 2020 – Jun 2020

EDUCATION

- Masters in Software Engineering – University of Amsterdam | 2025 - 2026
- Masters in Structural Engineering – BITS Pilani, India | 2018 - 2020 | 9.04 / 10.0

Notable Projects

- Order Execution System to trade on Deribit using C++17: Application for low-latency interaction, offering REST API access to crypto symbols and a WebSocket server for client subscriptions.
- VanillaVision - Twin Pricing Engine (C++17): Multi-threaded pricing engine enabling direct comparison of theoretical and simulation methods by evaluating runtime and accuracy.
- Black-Scholes Option Pricing Model Web App (Python): This project provides an interactive Black-Scholes Option Pricing Model Web App that hat visualizes Call and Put option prices on a dynamic heatmap.
- Ant Colony Structural Optimization Technique (ACO) using MATLAB and C programming
- Python Image Processing Application – DRDO, India using Python programming
- Other personal projects in C++17 and related web applications

Notable Certifications & Accolades

- Certifications in Mastering Advanced C++17 , Harvard CS50 , Meta full-stack development with JavaScript and React and Duke University – Java Programming in Coursera.
- MERIT 2022 winner (Global Competition conducted by ICE – UK)
- Scholarships: BITSAT Graduate Application Scholarship and PG scholarship from MHRD

SOFT SKILLS

- Highly adaptable and always hungry for knowledge.
- Strong analytical and debugging mindset
- Effective communicator and team collaborator, fluent in English (C1 level).